RIPHAH INTERNATIONAL UNIVERSITY, LAHORE

FINAL YEAR PROJECTS (FYP) GUIDELINES

Entrepreneur and Innovation-Category Projects

Operational Guidelines and Governance Policy

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Document Title	Operational Guidelines and Governance Policy for Entrepreneur and Innovation - Category Final Year Projects
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Policy precedence

This document shall be read together with applicable Riphah International University regulations, FYP procedures, academic-integrity requirements, HEC policies/frameworks, research-ethics requirements, intellectual-property provisions, data-protection obligations, and applicable law. Where a higher-authority requirement conflicts with this document, the higher-authority requirement shall prevail.

Version 1.0 — Fall 2026 Onwards

Contents

PART-A: Startup Ideation & Customer Discovery 2
Section 1: Idea Generation 2
Section 2: Customer Discovery & Fieldwork 2
Section 3: Startup Idea Validation 2
PART-B: Product Analysis, Design & Prototype 3
Section 4: Software Requirement Analysis Specification (SRS) 3
Section 5: System Architecture & Technical Design 3
Section 6: Prototype & MVP Development 4
PART-C: Business Plan & Strategy 4
Section 7: Business Model & Value Proposition Canvas 4
Section 8: Market & Competitor Analysis 4
Section 9: Marketing Plan & Go-To-Market Strategy 5
Section 10: Financial Plan & Projections 5
PART-D: Business Branding, Growth, Risk Management & Legal Issues 5
Section 11: Legal, Ethical & Regulatory Framework 5
Section 12: Risk Management Plan 6
Section 13: Brand Strategy & Visual Identity Guidelines 6
Section 14: Team Building & Human Capital Strategy 6
Section 15: Scaling & Growth Strategy 6
Appendices Guidelines 6
Recommended (Should Do) 7
Optional (Nice to Have) 7
Universal Dataset Governance & Data Disclaimer Policy 7
AI Tool Usage Policy: 8

PART-A: Startup Ideation & Customer Discovery

Section 1: Idea Generation

Students are expected to document the origin and evolution of their startup concept:

- Define primary/secondary users, outline research methods, and itemize core user pain points.
- Format the problem statement as *"[User] needs a way to [need] because [insight]"*.
- List all generated ideas, select the final concept, and provide the selection rationale.
- Assess technical feasibility, market demand, scalability, and choose an initial prototype medium.
- Test the concept with users, record feedback, and apply all 7 SCAMPER method dimensions.

Section 2: Customer Discovery & Fieldwork

Students must demonstrate direct customer engagement and empirical research:

- Define target demographics and formulate testable hypotheses on problem severity and willingness to pay.
- Prepare 5–10 open-ended questions and conduct qualitative interviews with 7–10 prospective customers.
- Consolidate customer feedback, document current workarounds, and confirm problem validation.
- Detail specific product adjustments and feature improvements made based on customer feedback.

Section 3: Startup Idea Validation

Students are required to validate the commercial and operational viability of their idea:

- Rate problem pain (1–10 scale), frequency of occurrence, and total market potential.
- Supply a log of 10+ customer interviews documenting interest levels and willingness to pay.
- Complete the UVP formula: *"We help [target customer] solve [problem] by providing [solution] unlike [alternative]"*.
- Specify the MVP format, pricing structure, key validation metrics, and proof of customer willingness to pay.
- Formally declare and justify a final Persevere, Pivot, or Drop strategic direction.

PART-B: Product Analysis, Design & Prototype

Section 4: Software Requirement Analysis Specification (SRS)

Students must author a formal Software Requirements Specification (SRS) document for their project:

- Define the product purpose, intended audience, boundaries, and referenced documentation.
- Document overall system context, user classes, operating environments, and system constraints
- List all explicit technical assumptions, operational dependencies, and system limits.
- Specify requirements for User Interfaces (GUI standards), Hardware, Software/APIs, and Communications.
- Detail functional requirements grouped by system feature with assigned priority levels (High/Medium/Low) and unique requirement codes (e.g., REQ-SF1-1). Provide complete Use Case models, actor profiles, and detailed step-by-step use case scenarios.
- Define non-functional requirements including measurable standards for performance, response times, scalability, availability, and reliability.
- Set measurable performance, security, reliability, and usability standards along with Use Case Models.
- Construct a Requirements Traceability Matrix linking requirements to software components and testing procedures.

Section 5: System Architecture & Technical Design

Students must present core software engineering diagrams illustrating system structure:

- Define the core architectural style (e.g., Client-Server, MVC, Microservices) and justify the selected technology stack.
- Draw a high-level system architecture diagram showing hardware components, software layers, and external third-party integrations.
- Map real-world conceptual entities, key attributes, and domain relationships using a Domain Model.
- Provide Sequence Diagrams illustrating object interactions, message passing, and execution flow for primary system features.
- Display database schemas, entity tables, primary/foreign keys, field types, and cardinalities via an Entity Relationship Diagram (ERD).
- Present an object-oriented Class Diagram detailing classes, attributes, methods, access modifiers, inheritance, and associations.

- Specify API interface contracts including REST/GraphQL endpoints, request parameters, payload structures, and error codes.
- Supply a Deployment Diagram detailing cloud infrastructure setup, web servers, database hosting, networks, and CI/CD pipelines.

Section 6: Prototype & MVP Development

Students must build, test, and evaluate a functioning Minimum Viable Product (MVP) or prototype:

- Choose low/high-fidelity prototype formats, prioritize core features, and justify the MVP design.
- Outline target user groups, testing methods, usability questions, and target success metrics.
- Document test execution, user feedback, key learnings, and resulting product iterations.
- Itemize prototype development costs and evaluate overall Problem-Solution fit.

PART-C: Business Plan & Strategy

Section 7: Business Model & Value Proposition Canvas

Students must present the strategic business architecture using standardized canvases:

- Map Customer Profile elements (Jobs, Pains, Gains) against Value Map features (Features, Relievers, Creators).
- Fill out all 9 standard BMC building blocks from Key Partners to Revenue Streams.

Section 8: Market & Competitor Analysis

Students are expected to evaluate external market dynamics, competitive forces, and market size:

- Evaluate Political, Economic, Social, and Technological drivers and their strategic implications.
- Rate competitive forces across Rivalry, New Entrants, Substitutes, Buyer Power, and Supplier Power.
- Identify internal Strengths/Weaknesses and external Opportunities/Threats with actionable strategies.
- Calculate market values using standard equations: $\text{TAM} = \text{Total Customers} \times \text{Annual Value}$.
- Cite primary/secondary research sources, analyze growth potential, and include visual charts.

Section 9: Marketing Plan & Go-To-Market Strategy

Students must construct a Go-To-Market (GTM) plan:

- Segment the market across standard criteria and build detailed target customer personas.
- Formulate Brand Purpose, Vision, Mission, Core Values, and the formal Positioning Statement.
- Analyze direct/indirect competitors and define the 4Ps (Product, Price, Place, Promotion).
- Map digital marketing tactics and the customer journey from Awareness to Retention.
- Outline the itemized marketing budget, key conversion metrics (CAC, LTV), and execution roadmap.

Section 10: Financial Plan & Projections

Students must formulate a complete financial model:

- Itemize one-time Capital Expenditures (CapEx) and recurring monthly Operating Expenses (OpEx).
- Build a Work Breakdown Structure (WBS) and calculate development costs using COCOMO models.
- Establish pricing structures and build 1-year monthly and 5-year annual revenue forecasts.
- Provide a 12-month Cash Flow Statement and calculate Net Burn Rate and Cash Runway.
- Calculate break-even volume using: $\text{Break-Even Units} = \frac{\text{Fixed Costs}}{\text{Price per Unit} - \text{Variable Cost per Unit}}$.
- Define funding needs, equity splits, exit strategy, and unit metrics (LTV, CAC, LTV/CAC ratio).

PART-D: Business Branding, Growth, Risk Management & Legal Issues

Section 11: Legal, Ethical & Regulatory Framework

Students must address legal compliance and ethical standards:

- Select and justify a business structure (Sole Proprietorship, Partnership, SMC, LLC).
- Detail IP protection plans covering Patents, Trademarks, Copyrights, and Trade Secrets.
- Outline equity vesting, decision rights, exit terms, and data privacy/ethical compliance.
- Complete a legal checklist covering business registration, NDAs, user contracts, and tax filing.

Section 12: Risk Management Plan

Students must proactively manage potential venture risks:

- Categorize risks (Financial, Market, Technical, Legal, Operational) and calculate Risk Scores (Probability × Impact).
- Plot a 3x3 Risk Matrix, define explicit mitigation strategies (Avoid, Reduce, Transfer, Accept), and backup contingency plans.

Section 13: Brand Strategy & Visual Identity Guidelines

Students must define their corporate branding identity:

- Document Brand Purpose, Vision, Mission, Messaging Pillars, Tagline, and Elevator Pitch.
- Define logo usage rules, color palettes, typography specs, and overall Brand Book guidelines.
- List deliverables for web/app UI assets, marketing collateral, packaging, and CX touchpoints.

Section 14: Team Building & Human Capital Strategy

Students must outline organizational strategy and human resources planning:

- Define C-level team roles, responsibilities, and Hire vs. Outsource decisions.
- Provide a 6–12 month recruitment plan, company culture values, selection criteria, and employee KPIs.
- Document equity split terms, vesting schedules, and conflict resolution protocols.

Section 15: Scaling & Growth Strategy

Students must deliver a roadmap for long-term venture growth:

- Select and justify a primary expansion strategy (Penetration, Expansion, Development, Diversification).
- Outline target market expansion tactics, system automation tools, and strategic partnership models.
- Track key scaling metrics (CAC, LTV, Churn, Burn) and construct a 0–12 month phased growth timeline.

Appendices Guidelines

- Include full customer interview questionnaire scripts and raw participant response data.

- Attach secondary market research data tables, industry reports, and citation excerpts.
- Provide high-resolution screenshots of UI wireframes, mockups, and prototype implementations.

Recommended (Should Do)

- **Trust & Verification Mechanisms:** Implement clear verification workflows, vendor profiling, secure payment modules, and rating/feedback loops to resolve user trust issues.
- **Realistic Financial Costing:** Detail realistic cost estimates for development, cloud hosting, API dependencies, marketing, and break-even timelines.
- **User Survey & Field Validation:** Gather empirical requirements through field interviews, vendor engagement, and target audience surveys.

Optional (Nice to Have)

- Demonstrate early customer traction metrics (e.g., landing page signups, pilot feedback, or letters of intent).
- Formulate IP/trademark protection strategies and multi-year financial scaling projections.

Universal Dataset Governance & Data Disclaimer Policy

(Applies to all Final Year Projects across all categories using datasets, data collection, or processing)

- **Data Anonymization:** All Personally Identifiable Information (PII)—including names, contact details, email addresses, national ID numbers (CNIC), location metadata, and facial images—must be thoroughly masked, scrubbed, or anonymized prior to model training, processing, or repository submission.
- **Compliance with Data Laws:** Datasets must strictly comply with local and international privacy laws (e.g., Pakistan Prevention of Electronic Crimes Act - PECA, GDPR). Unauthorized scraping or scraping from restricted/private portals without written consent is strictly prohibited.
- **Dataset Sourcing & Licensing:** Ensure all external datasets possess clear open-source or academic licenses (e.g., MIT, CC BY 4.0). Any domain-specific data (e.g.,

medical, legal, financial) collected directly from users requires institutional review, formal consent, or SME authorization.

- **Regional Context & Origin:** Datasets must be context-relevant and address local/origin-oriented requirements (e.g., Pakistani demographic or localized domain data) where applicable. Explicitly detail how data collection and availability challenges will be resolved.

AI Tool Usage Policy:

Students can extensively use AI for product development and documentation.

Please consult with your respective supervisors to address these points in your next documentation revision.